



Analysis of seasonal and annual rainfall trends in the northern region of Bangladesh



Sheikh Hefzul Bari^{a,*}, M. Tauhid Ur Rahman^{b,c}, Muhammad Azizul Hoque^b, Md. Manjurul Hussain^b

^a Department of Civil Engineering, Leading University, Sylhet, Bangladesh

^b Department of Civil and Environmental Engineering, Shahjalal University of Science and Technology, Sylhet 3114, Bangladesh

^c Climate Lab, HEQEP, Dept. of Civil Engineering, Military Institute of Science and Technology, Dhaka-1216, Bangladesh

ARTICLE INFO

Article history:

Received 10 November 2015

Received in revised form 7 February 2016

Accepted 10 February 2016

Available online 19 February 2016

Keywords:

Seasonal rainfall

Trend analysis

Mann–Kendall test

The sequential Mann–Kendall test

Sen's slope

Northern Bangladesh

ABSTRACT

The aim of the present study was to investigate 50 years (1964–2013) of seasonal and annual rainfall trends and their fluctuation over time in northern Bangladesh. After testing the autocorrelation, non-parametric Mann–Kendall test along with Sen Slope estimator was used to examine rainfall trends and their magnitudes. The sequential Mann–Kendall test was used to identify any fluctuations in the trends over time and to detect the possible points of change in the rainfall series. We found that pre-monsoon and post-monsoon rainfall is increasing in most of the rainfall stations. The only decrement in pre-monsoon rainfall was found for Ishurdi (1.28 mm/year). However, the sequential Mann–Kendall test detected decreasing pre-monsoon rainfall trend after early the 1990s. Monsoon rainfall showed a decreasing trend in the majority of the area studied. The maximum decrement in monsoon rainfall was found for Sylhet station (8.10 mm/year) and minimum in Mymensingh (1.53 mm/year). An upward monsoon rainfall trend was found for Rangpur (2.02 mm/year). Annual rainfall followed the monsoon rainfall trend. However, all of the positive and negative trends were found statistically non-significant at 95% confidence limit with the only exception for monsoon and annual rainfall at Rajshahi station. Rajshahi station was the only region where the monsoon and annual rainfall has a significant negative trend at 95% confidence limit. The sequential Mann–Kendall test detected several non-significant points of change for seasonal and annual rainfall at most of the stations. Periodic fluctuations were also detected. We observed that there were decreasing seasonal rainfall trend after early the 1990s for the majority of the stations.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

Scientific evidence shows that climate change has begun to manifest itself, globally, in the form of increased downpours and storms, rising temperature and sea level, and retreating glaciers, among others. The increasing sea surface temperature, especially, has a substantial effect on the increasing rainfall (Trenberth, 2011). Various research studies indicate that global warming has reformed precipitation resulting in frequent extreme weather events and risks (Briffa et al., 2009; Vasiliades et al., 2009; Zhang et al., 2009). According to IPCC (Intergovernmental Panel on Climate Change) report (McCarthy, 2001), global precipitation is subjected to increase. However, an increment or decrement in the regional scale is not denied (McCarthy, 2001). It was seen that global monsoon precipitation had a tendency to increase from 1901 to 1955 and a decreasing trend since the 1950s (Zhang and Zhou, 2011).

Geographical settings and hydrological uncertainty make Bangladesh one of the top climate susceptible countries. Recent studies show that the climate of Bangladesh has changed (Shahid, 2010a; Shahid and Khairulmaini, 2009). For Bangladesh, a few studies were carried out

on rainfall pattern (Ahmed and Kim, 2003; Islam and Uyeda, 2008; Ahammed et al., 2014) and trend (Immerzeel, 2008; Shahid, 2011; Hasan et al., 2014). However, they do not potentially signify the seasonal rainfall trend and their fluctuation over time for different micro-climatic regions of Bangladesh, especially for the northern region.

Ahammed et al. (2014) studied the variability and return period of the annual daily maximum rainfall for Dhaka, Bangladesh. Hasan et al. (2014) studied the rainfall trend for the southeast part of coastal Bangladesh and found increasing pre-monsoon rainfall trend. Nowreen et al. (2015) reported that frequency of rainy days has reduced at northeastern Bangladesh. Rahman et al. (2015) found a decreasing monthly rainfall trend for Rajshahi station that is west subzone. They also studied Bogra and Sylhet station but found no trend in monthly rainfall. Study on neighboring Indian regions reveal weaker monsoon rainfall (Basistha et al., 2009; Choudhury et al., 2012). However, no research was found for northern Bangladesh in seasonal scale. This part of Bangladesh is representative of most of the climatic subdivisions of the country and covers the wettest and the driest places. The area consists of most of the agronomic land of the country and agriculture is mostly rain-fed. It is therefore vital to understand the changes in rainfall regime in this area. Change in rainfall trends and their spatial distribution may modify the natural stream discharges, fresh water availability, groundwater recharge, etc., causing an interruption in agricultural production.

* Corresponding author.

E-mail address: bari_cee1989@yahoo.com (S.H. Bari).

Even though various statistical methods are now frequently used by researchers for analysing trends, their fluctuations, and change point detection (Nasri and Modarres, 2009; Farhana and Rahman, 2011; Afzal et al., 2011; Shahid, 2011; Patra et al., 2012; Martinez et al., 2012; Sonali and Kumar, 2013; Wang et al., 2013; Jha and Singh, 2013), non-parametric methods are preferred over parametric ones. It is mainly because parametric tests are mostly appropriate for normally distributed data. If the assumed data distribution departs considerably from the standard, then the non-parametric test gives much better results than the parametric one

(Hirsch et al., 1992). It is seen that many hydrologic data series are either positively skewed or are not always normally distributed (Hirsch et al., 1992; Siddik and Rahman, 2014).

In this study, an analysis of rainfall trends and their fluctuations over time on seasonal and annual scale are done for the northern part of Bangladesh. The possible causes of rainfall changes are also discussed. The non-parametric Mann–Kendall test was used to detect trends; Sen's Slope was applied to identify trend magnitude and sequential Mann–Kendall test to find trend fluctuations over time and abrupt changes.

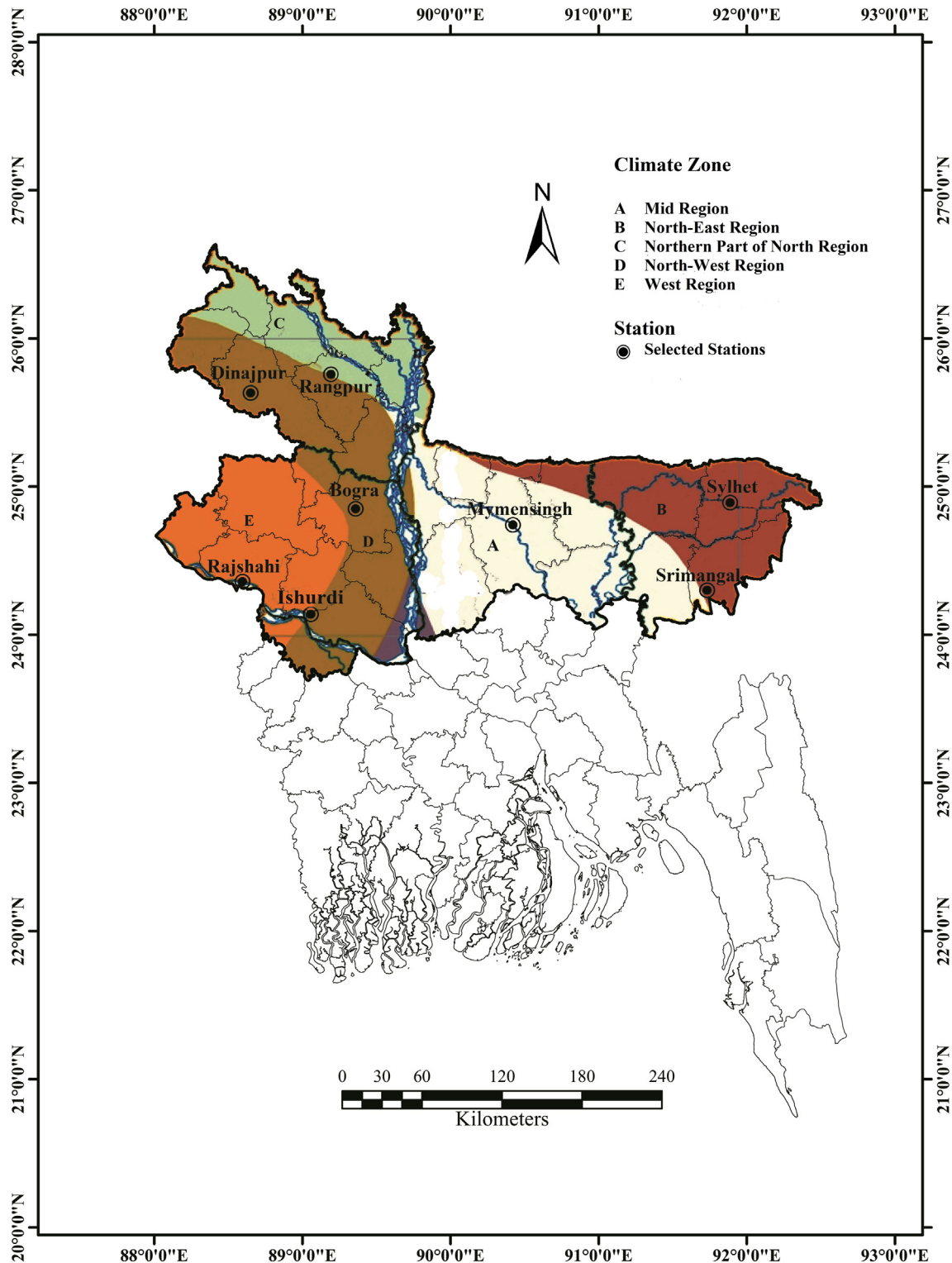


Fig. 1. Location map of northern Bangladesh with rain gauge station locations (modified after Rashid, 1991). The bold line represents the study area boundary.

Autocorrelation of the data was tested afore applying the Mann–Kendall trend test to obtain accurate results. In addition, Kendall's tau (τ) was also used to identify the strength of the monotonic trend.

2. Study area

Bangladesh is located at the crossroads of the South Asian monsoon circulation, which is at the entry and exit points of the Bay of Bengal Branch of the South Asian Monsoon (Ahmed and Karmakar, 1993; Ahmed, 1994). There are widespread differences in the intensity of the seasons at different places within the country. For example, the summer monsoon prevails from early June to mid-October, with an average duration of 110 days in the west to 134 days in the southeast and an average of 60 rain days in the west to 100 days in the northeast and southeast (Ahmed and Kim, 2003). For the present study, we focused only on the northern part of Bangladesh. The location map of the study area is shown in Fig. 1. This region covers the four climatic sub-zones (Zone B, C, D, and E) and part of zone G (Fig. 1) out of seven climatic zones, including the two most distinct climatic regions, namely, the north-west and the northeast regions. While the mean annual rainfall of Bangladesh is about 2300 mm, the average annual rainfall in the northwest region is 1329–1700 mm in contrast to the northeast part of the country, which has 4101–4338 mm rainfall (Shahid, 2008; Shahid and Behrawan, 2008). The description of the different climate sub-zones included in the study can be found elsewhere (e.g. Rashid, 1991).

3. Materials and methods

Monthly rainfall time series from 1961 to 2013 was collected from Bangladesh Agricultural Research Council (BARC). BARC, in turn, collects rainfall data from Bangladesh Meteorological Department (BMD). The location of BMD climate stations available in the study area is shown in Fig. 1.

To ensure the quality of rainfall time series, the following check (Siddik and Rahman, 2014) was applied: (a) a year was considered complete if the entire month's rainfall was available and (b) a station series was considered complete if it had equalled to or more than 80% complete years. While assessing data quality, inconsistency was found in the years 1961 and 1962 in Srimangal. Hence, only rainfall data for the period of 1964 to 2013 were used in the analysis. The monthly rainfall data were used to formulate annual and seasonal time series. The four standard seasons in Bangladesh are winter (dry): December to February; pre-monsoon: March to May; monsoon: June to September; and post-monsoon: October to November (Siddik and Rahman, 2014).

The monotonic trend of rainfall time series was measured using the Mann–Kendall trend test (Mann, 1945). However, this test is based on the assumption of uncorrelated data. If any serial correlation exists, the Modified Mann–Kendall (MMK) test (Hamed and Rao, 1998) should be used. Before starting a monotonic trend test, lag 1 autocorrelation for all the time series were checked using the autocorrelation function (ACF). A negative or positive value of the lag 1 autocorrelation would indicate the presence or absence of the biennial oscillation, respectively (Kripalani et al., 2007). However, no time series was significantly serially correlated in 95% confidence limit. Hence, the Mann–Kendall test was applied. The Mann–Kendall statistics S is given as follows (Sonali and Kumar, 2013):

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sign}(x_j - x_k)$$

where

$$\text{Sign}(x_j - x_k) = \begin{cases} +1 & \text{if } (x_j - x_k) > 0 \\ -1 & \text{if } (x_j - x_k) < 0 \\ 0 & \text{if } (x_j - x_k) = 0 \end{cases}$$

and x_k and x_j are the sequential data values; n is the length of the data set.

The variance of S is given by

$$\text{VAR}(S) = \frac{1}{18} [n(n-1)(2n+5)]$$

If tie is present in the data, then variance of S is given by

$$\text{VAR}(S) = \frac{1}{18} \left[n(n-1)(2n+5) - \sum_{i=1}^n t_i(i-1)(2i+5) \right]$$

where t_i is the number of ties of extend i .

A positive (negative) value of S indicates a positive (negative) trend. The significance of trend was determined at 95% confidence limit ($\alpha = 0.05$).

In the above test, another very useful index is the Sen's slope, which is the magnitude of the monotonic trend and is given as follows (Sen, 1968):

$$\beta = \text{Median} \left(\frac{x_i - x_j}{i - j} \right), \text{ for all } j < i$$

Tau (Kendall, 1938, 1948) measures the strength of the monotonic relationship between x and y . Kendall's tau correlation coefficient is given by

$$\tau = \frac{S}{n(n-1)/2}$$

Table 1

Details of the Mann–Kendall trend analysis.

Station name	Variable	Kendall's tau	Kendall's S	p-value (two-tailed)	Sen's slope
Bogra	Winter	0.02	29.00	0.814	0.00
	Pre-monsoon	0.06	69.00	0.569	0.70
	Monsoon	−0.09	−114.00	0.345	−2.55
	Post-monsoon	0.09	113.00	0.349	0.82
	Annual	−0.02	−19.00	0.880	−0.62
Dinajpur	Winter	0.19	226.00	0.059	0.25
	Pre-monsoon	0.17	210.00	0.080	1.50
	Monsoon	0.01	18.00	0.887	0.00
	Post-monsoon	0.04	49.00	0.687	0.29
	Annual	0.10	121.00	0.314	3.55
Ishurdi	Winter	0.08	102.00	0.397	0.13
	Pre-monsoon	−0.10	−124.00	0.303	−1.28
	Monsoon	−0.15	−185.00	0.124	−4.91
	Post-monsoon	0.00	5.00	0.973	0.00
	Annual	−0.11	−137.00	0.255	−5.43
Mymensingh	Winter	0.03	31.00	0.802	0.04
	Pre-monsoon	0.07	89.00	0.462	1.27
	Monsoon	−0.03	−37.00	0.763	−1.53
	Post-monsoon	0.04	54.00	0.658	0.70
	Annual	0.00	−3.00	0.987	−0.04
Rajshahi	Winter	−0.07	−83.00	0.4922	−0.15
	Pre-monsoon	0.05	55.00	0.6514	0.37
	Monsoon	−0.25	−307.00	0.0105	−7.33
	Post-monsoon	−0.03	−39.00	0.7506	−0.25
	Annual	−0.20	−240.00	0.0456	−7.24
Rangpur	Winter	0.07	85.00	0.4817	0.08
	Pre-monsoon	0.12	149.00	0.2157	1.67
	Monsoon	0.09	114.00	0.3445	2.02
	Post-monsoon	0.15	178.00	0.1387	1.78
	Annual	0.18	223.00	0.0633	8.42
Sylhet	Winter	−0.08	−95.00	0.4315	−0.22
	Pre-monsoon	0.09	115.00	0.3403	2.94
	Monsoon	−0.17	−214.00	0.0748	−8.10
	Post-monsoon	−0.05	−65.00	0.5924	−0.65
	Annual	−0.09	−116.00	0.3361	−6.24
Srimangal	Winter	−0.14	−173.00	0.1500	−0.40
	Pre-monsoon	0.09	107.00	0.3752	1.87
	Monsoon	−0.17	−206.00	0.0864	−3.70
	Post-monsoon	0.05	65.00	0.5924	0.53
	Annual	−0.08	−102.00	0.3982	−2.78

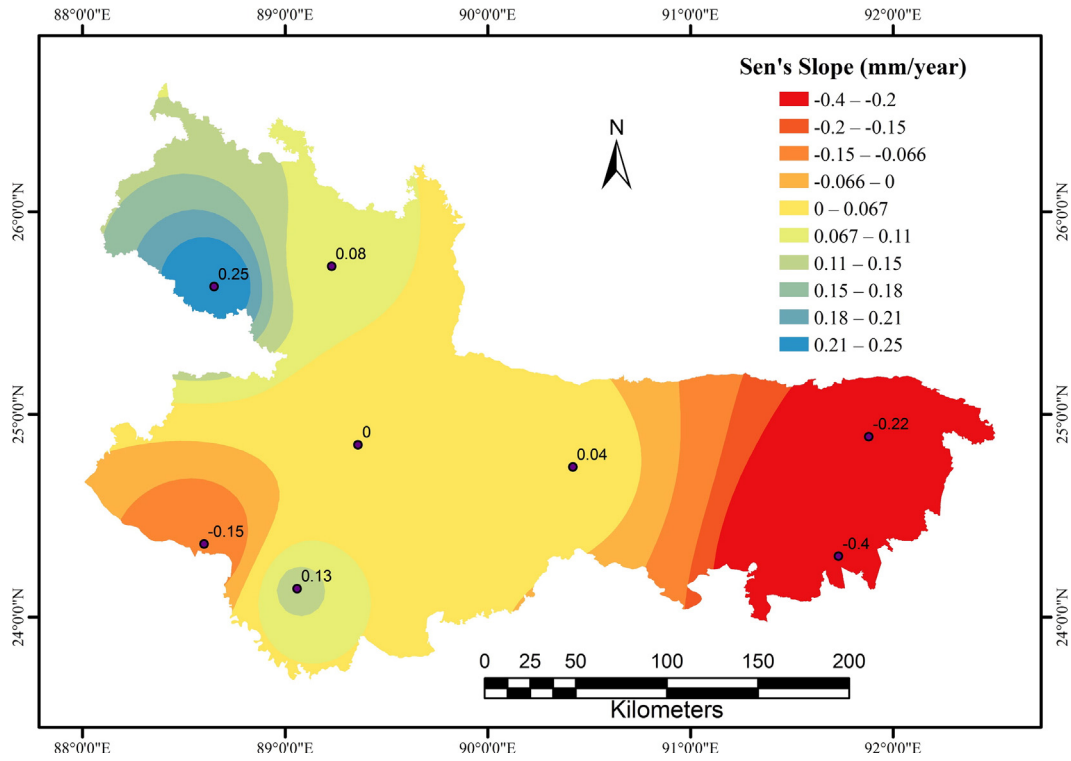


Fig. 2. Spatial distribution of trend magnitude (mm/year) for winter rainfall.

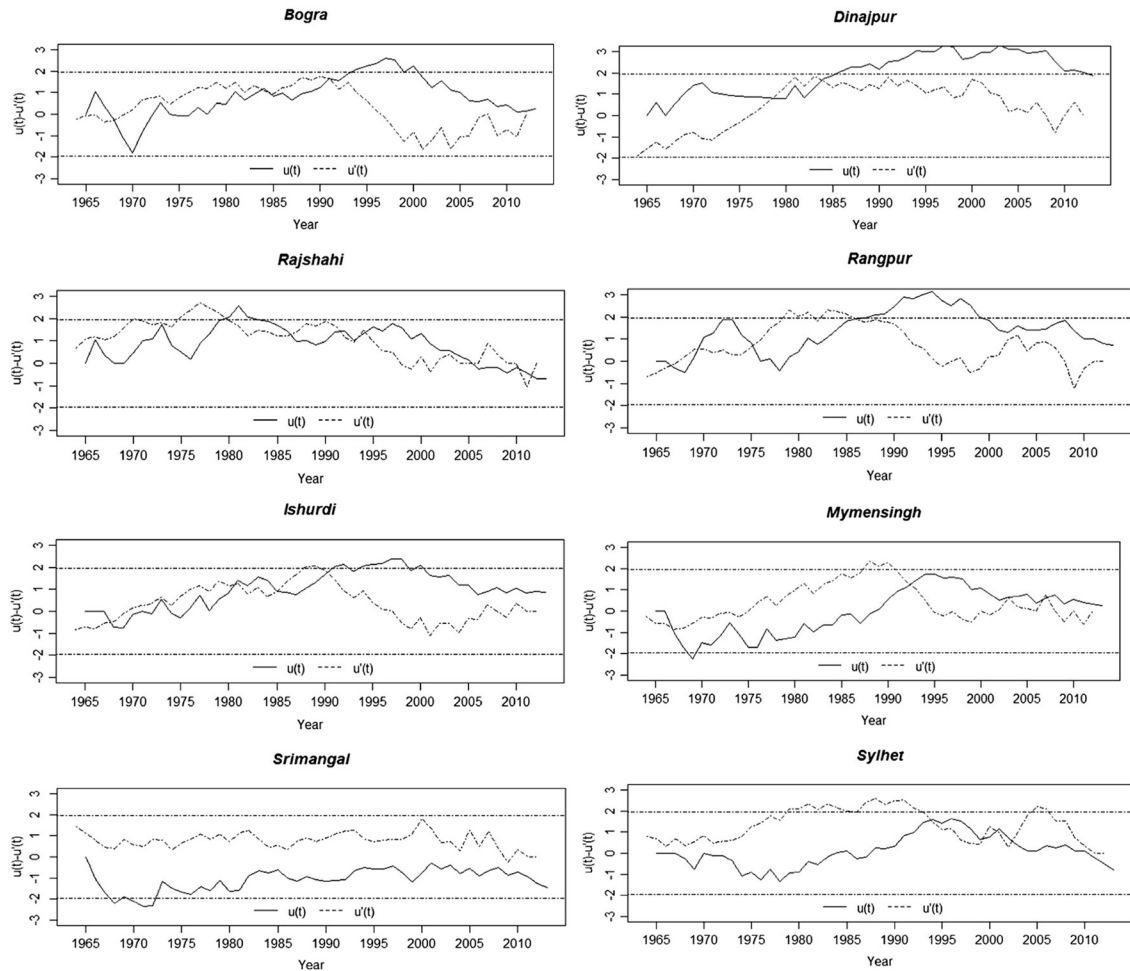


Fig. 3. Sequential Mann–Kendall test rank statistics for winter rainfall. Dotted horizontal straight lines indicate the lower limit and upper limit of 95% confidence interval.

The sequential Mann–Kendall (SQ-MK) test was used to determine the approximate year of the start of a significant trend (Lu et al., 2004; Partal and Kahya, 2006; Some'e et al., 2012; Tabari et al., 2012). This test also shows fluctuation in the trend over time. This test sets up two series, a progressive one $u(t)$ and a backward one $u'(t)$. If they cross each other and diverge beyond the specific threshold value (95% confidence limit in the present analysis), then there is a statistically significant change point. The point where they cross each other indicates the approximate year at which the trend begins (Mosmann et al., 2004). The following steps were applied to calculate $u(t)$ and $u'(t)$ (Nasri and Modarres, 2009):

1. The values of x_j annual mean time series ($j = 1 \dots n$) were compared to x_i ($i = 1, \dots, j - 1$). For each comparison, the number of cases $x_j > x_i$ was counted and denoted by n_j .
2. The test statistics t was then calculated by the equation

$$t_j = \sum_{i=1}^j n_j$$

3. The mean and variance of the test statistics are

$$E(t) = \frac{n(n-1)}{4} \text{ and } \text{Var}(t_j) = [j(j-1)(2j+5)]/72$$

4. The sequential values of the statistics $u(t)$ were then calculated as

$$u(t) = \frac{t_j - E(t)}{\sqrt{\text{Var}(t_j)}}$$

Similarly, the values of $u'(t)$ were computed backward, starting from the end of the series.

The spatial distribution of trend magnitude (by Sen Slope) was estimated using the inverse distance weighting (IDW). IDW is a widely used quick deterministic interpolator that is exact and require no assumption for data (Karimi, 2009; Longley, 2005). Further reading for this method can be found elsewhere (Brooker, 1991; Fortin and Dale, 2005; Lloyd, 2010; Longley, 2005).

6. Results

The Mann–Kendall test statistics are presented in Table 1. The table shows that monsoon and the annual rainfall of Rajshahi station have a significant negative trend in 95% confidence limit. The rest of the stations have a non-significant trend (either positive or negative) in 95% confidence limit. A detailed description of seasonal and annual rainfall trend analysis result is given in the following paragraphs.

Rainfall is scarce in the country during the winter. It is found that the amount of winter rainfall is decreasing in western and northeast climatic regions (area of extremes) of Bangladesh. An unchanging winter rainfall was found for Bogra (0 mm/year). Rest of the stations show a diminutive magnitude of positive trend. All of the positive and negative trends were found statistically non-significant at 95% confidence limit. Besides, relationship (measured by Kendall's tau value) of monotonic trend with time was found weak. The spatial distribution of trend magnitude (mm/year) for winter is shown in Fig. 2. Results from the sequential Mann–Kendall test reveal more than one change point for all stations except for Srimangal. A typical scenario for majority of the stations is that there were change points between 1980 and 1990. The $u(t)$ statistics depicts that, after a change in 1967, winter rainfall in Mymensingh leaned toward an increasing trend. A more or less similar trend was observed in Ishurdi, Bogra, Dinajpur, and Rangpur. Although several change points were detected for each station, they were statistically non-significant at 95% confidence limit. Both $u(t)$ and $u'(t)$ statistics remained almost parallel for Srimangal rainfall station in winter,

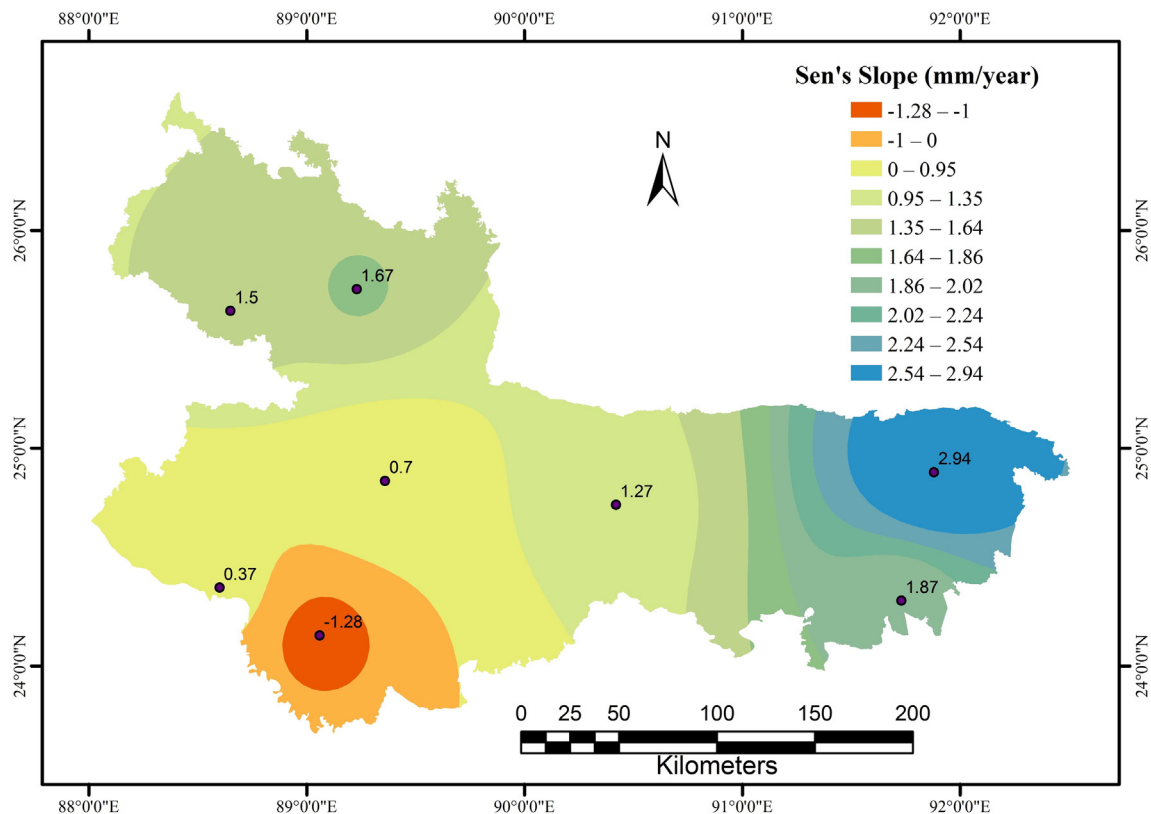


Fig. 4. Spatial distribution of trend magnitude (mm/year) for pre-monsoon rainfall.

indicating no abrupt change. The graphical representation of sequential Mann–Kendall test statistics is shown in Fig. 3.

Pre-monsoon rainfall shows a non-significant positive trend (at 95% confidence limit) for all the rain gauge stations except for Ishurdi. The increment in pre-monsoon rainfall is greater in low-lying northeastern part of Bangladesh with a maximum value of 2.93 mm/year at Sylhet.

By contrast, the increment in pre-monsoon rainfall is lowest for Bogra. Fig. 4 shows the spatial distribution of the trend magnitude of pre-monsoon rainfall in the study area. Although increment and decrement in pre-monsoon rainfall were observed, Kendall's tau reveals a weak relationship with time. $u(t)$ and $u'(t)$ statistics of the sequential Mann–Kendall test (Fig. 5) experienced more than one non-significant change point for all stations. A significant change point was identified for Bogra in 2003 and Ishurdi in 1978, 1980, and 1981. An increasing trend was detected for Mymensingh, Srimangal, Dinajpur, and Sylhet up to the early 1990s. After that, a slightly decreasing trend was observed at these stations. At Sylhet, the progressive and retrograde statistics progressed almost parallel after the 1980s. The common fact is that nearly all stations showed decreasing trend after the 1990s.

Monsoon in northern Bangladesh unveils a decreasing trend for all observation stations except Dinajpur (Fig. 6). The highest decrement in monsoon rainfall for northeastern part of Bangladesh was found for Sylhet (about 8.10 mm/year), while in the west region, Rajshahi had the greatest decrement (about 7.33 mm/year). Although the sequential Mann–Kendall test shows a decreasing trend for most of the measuring

stations (Fig. 7), a consecutive periodic change cannot be neglected for some stations. Considering $u(t)$ statistics, a decreasing trend of monsoon rainfall was found for Bogra, Rajshahi, Ishurdi, Srimangal, and Sylhet. No significant change points were detected for them. Non-significant change point was found for Ishurdi (at 1977 and 1980) and Sylhet (at 1994). Meanwhile, the progressive and retrograde statistics remained almost parallel for Rajshahi. Periodic fluctuations in monsoon rainfall were found in Mymensingh, Dinajpur, and Rangpur. In Mymensingh, a decreasing trend in monsoon rainfall up to the year 1975 was observed followed by an increasing rainfall pattern till early 1995. After early 1995, the monsoon rainfall appeared to drop for Mymensingh. A periodic increasing and decreasing trend followed by two possible significant change point between 1993 and 2003 were observed for Dinajpur. A significant change point was found in Rangpur at 1984. The increasing trend at Rangpur tended to decrease after 1990. It was found that decreasing monsoon rainfall trend is dominant after the 1990s for all stations.

Post-monsoon rainfall appeared to decrease in Rajshahi and Sylhet. However, their magnitudes were lower than unity. All the other stations exhibited increasing rainfall in post-monsoon season. The highest increasing rate was found in Rangpur (1.78 mm/year), while others were minuscule (less than 1 mm/year). A neutral trend was found for Ishurdi (0 mm/year). However, all of the positive and negative trends were found non-significant at 95% confidence limit. Kendall's tau also showed no strong relation of trend with time for any of the stations.

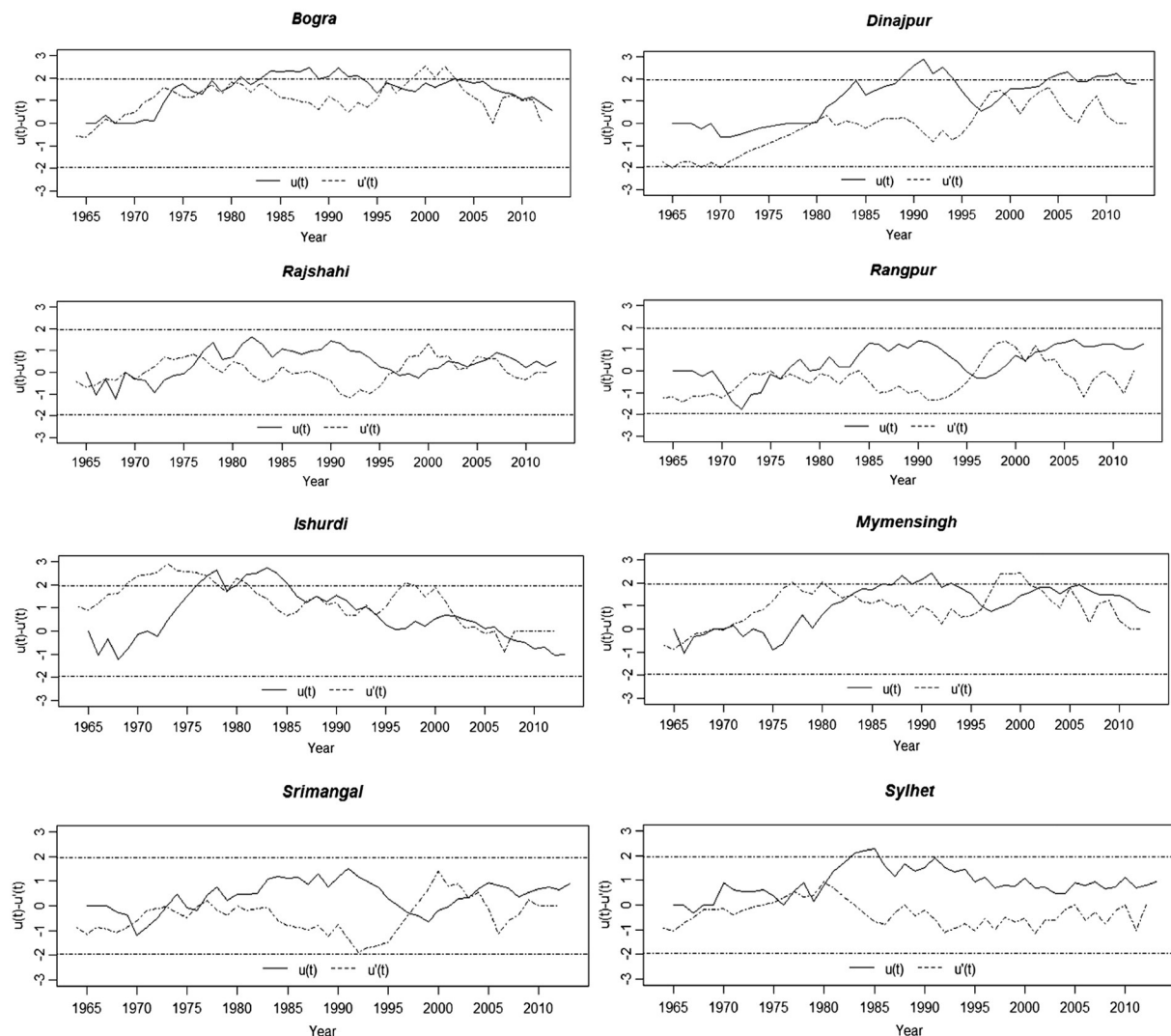


Fig. 5. Sequential Mann–Kendall test rank statistics for pre-monsoon rainfall. Dotted horizontal straight lines indicate the lower limit and upper limit of 95% confidence interval.

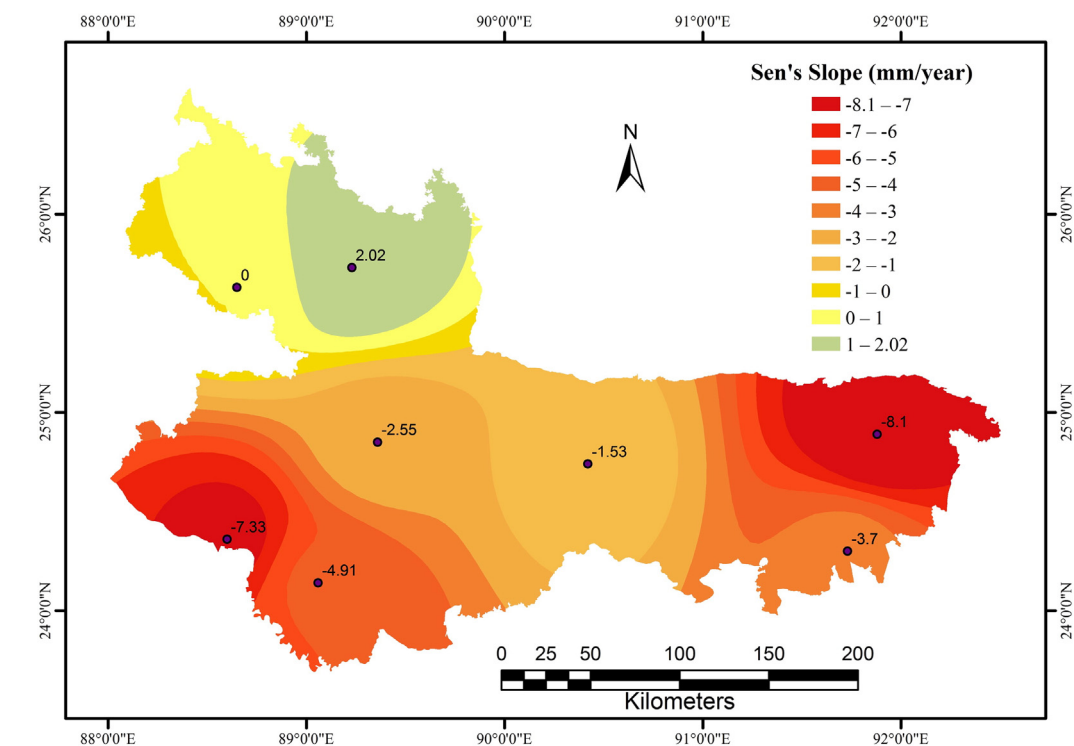


Fig. 6. Spatial distribution of trend magnitude (mm/year) for the monsoon rainfall.

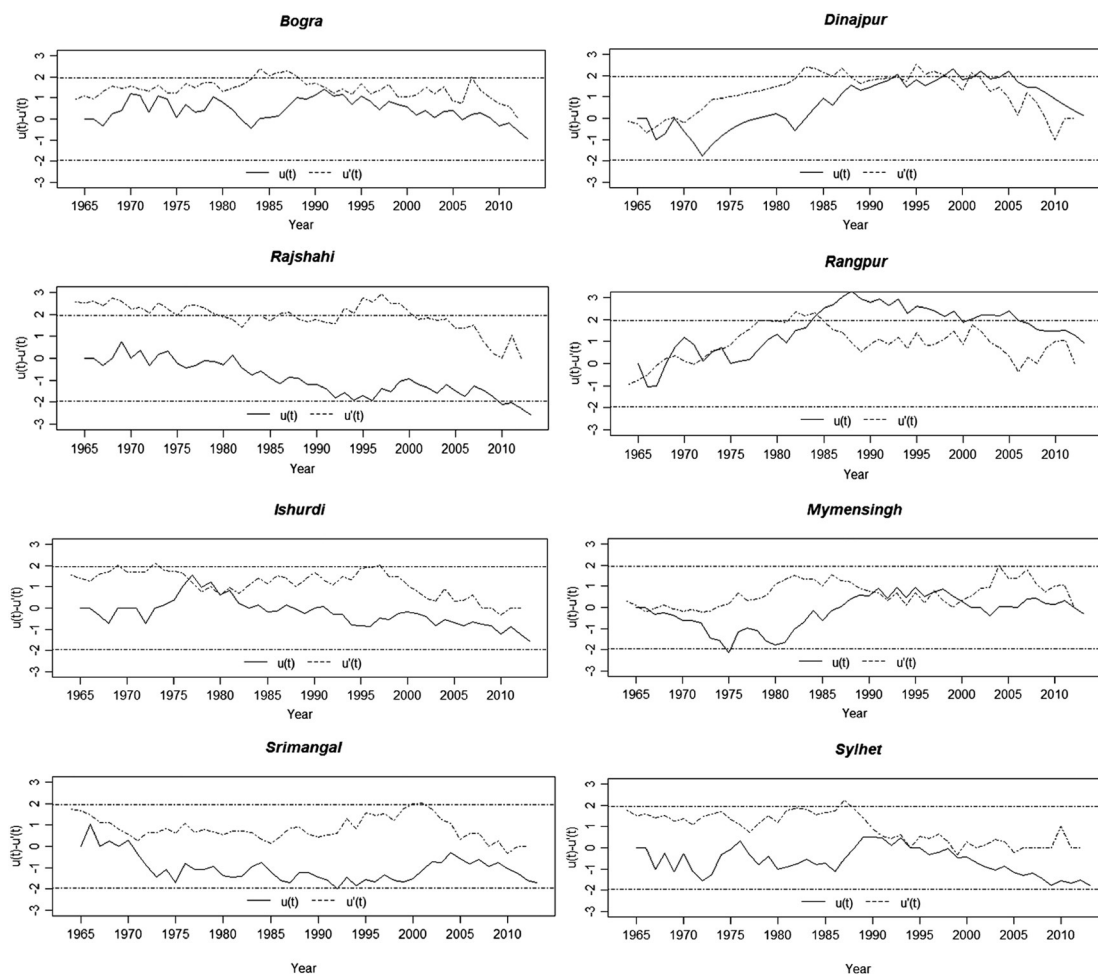


Fig. 7. Sequential Mann-Kendall test rank Statistics for monsoon rainfall. Dotted horizontal straight lines indicate the lower limit and upper limit of 95% confidence interval.

The spatial distribution of post-monsoon rainfall trend magnitude is shown in Fig. 8.

Post-monsoon rainfall shows no significant change points. Although every station experienced more than one non-significant change point, $u(t)$ only crossed the 95% confidence limit at Rangpur in early 2005. The results interpret that apparently there is no significant trend at all. The very small magnitude of Kendall's tau and Sen Slope values support this decision. The graphical representations of sequential Mann–Kendall test results are shown in Fig. 9.

Annual rainfall in northern Bangladesh showed negative trends except for Rangpur and Dinajpur. Rangpur and Dinajpur showed an increasing annual rainfall trend of about 8.42 mm/year and 3.55 mm/year, respectively. More precisely, the northwestern climatic region of Bangladesh experienced a positive rainfall trend whereas the rest experienced a negative trend. The highest negative trend was found in Rajshahi (7.24 mm/year). It is clear that northeast and west climatic region of Bangladesh are facing a higher rainfall decrement compared to other zones. The spatial distribution of annual rainfall trend magnitude is shown in Fig. 10.

The sequential version of the Mann–Kendall test depicts a variety of findings for different stations (Fig. 11). The $u(t)$ statistic represented a continuous decreasing trend of annual rainfall at Rajshahi. However, no change point was detected here. A consecutive increasing and decreasing trend with a significant change point in 1988 were observed for Bogra. Two change points in 1972 and 1988 were detected in Dinajpur. After a change point in 1988, the annual rainfall seemed to change intermittently for Dinajpur. Significant abrupt change was also detected in Rangpur after 1982. Consecutive increasing and decreasing trend and vice versa were found for Ishurdi and Mymensingh respectively. Two non-significant change points at a very short distance were found for Ishurdi. Likewise, three non-significant change points were observed in Mymensingh. The $u(t)$ statistics revealed an upward change after 1975 till 1990 in Mymensingh. Non-significant change points were also detected for Srimangal and Sylhet.

7. Discussion

Winter rainfall shows no significant change in northern Bangladesh. These findings of the study are analogous to other studies (Agrawala et al., 2003; Shahid, 2010a). Post-monsoon rainfall was found to be increasing non-significantly that is similar to the results of other sources (Agrawala et al., 2003; Shahid, 2010b) also.

The Mann–Kendall test identified an increasing pre-monsoon rainfall trend in Bangladesh. However, the sequential Mann–Kendall test identified a decreasing trend after the 1990s. The pre-monsoon rainfall in Bangladesh is governed by thunderstorms activities (Sanderson and Ahmed, 1979; Yamane et al., 2013). The thunderstorm season begins in the northeastern and eastern parts of the country by the first week of March. The activity of the thunderstorms during the pre-monsoon season depends upon the supply of moist air from the Bay of Bengal. Observation of thunderstorms in northern Bangladesh (Karmakar et al., 2015) supports the recent pre-monsoon rainfall trend behaviour found in this study.

Monsoon rainfall in northern Bangladesh is decreasing in most of the stations. This finding seems little contradictory with the results obtained by Shahid (Shahid, 2010b). Possible variation in the monsoon trend results for the current study from that of Shahid (2010a, 2010b) may be due to different data period (1957 to 2007). The present study covers the most recent years (1964–2013). However, several other studies in the same area and nearby Indian territories accede with the present study. Ahasan et al. (2010) found that the country's average monsoon rainfall is decreasing at a rate of 0.53 mm/year. Choudhury et al. (2013) reported decreasing monsoon rainfall for Sylhet region in recent time periods. Decreasing monsoon rainfall report was also found for neighboring states of India. Choudhury et al. (2012) reported that contribution of monsoon rainfall in total annual rainfall of monsoon months declined marginally at the rate of 1.70 mm while pre-monsoon and post-monsoon months increased non-significantly at an annual rate of 3.18 mm and 1.16 mm, respectively in Meghalaya. Stephenson

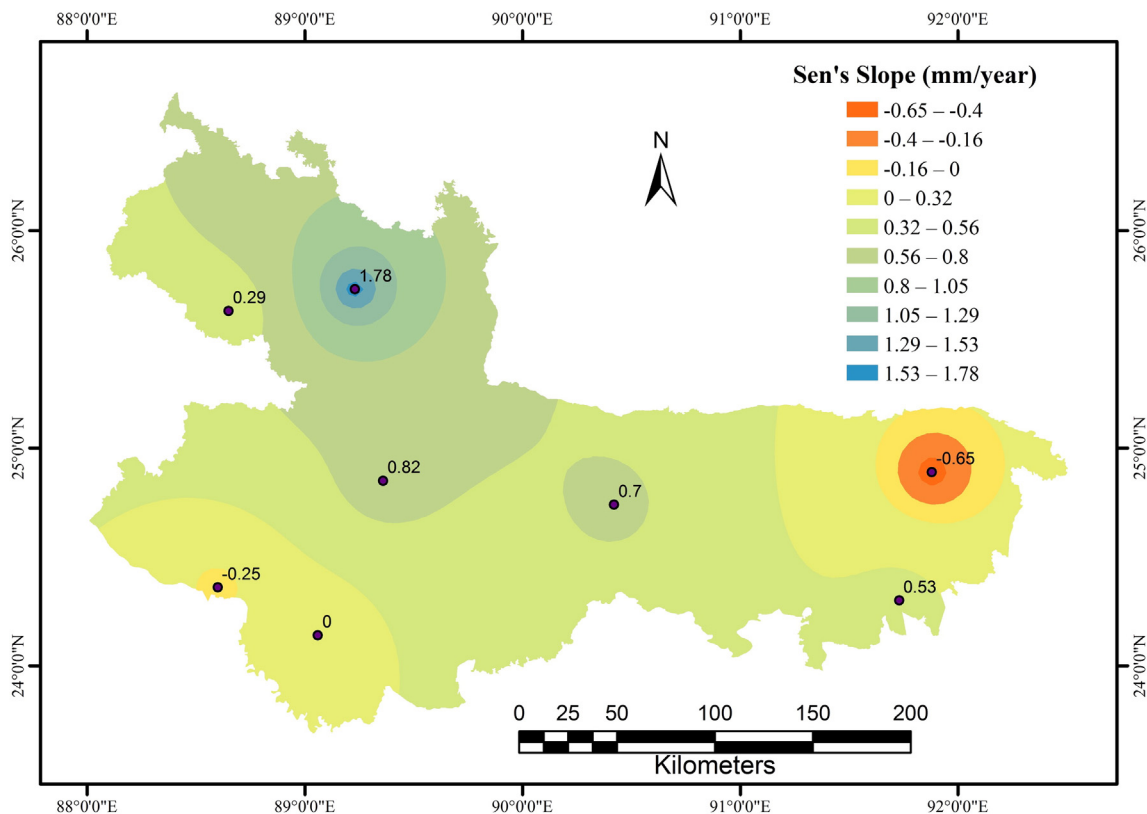


Fig. 8. Spatial distribution of trend magnitude (mm/year) for post-monsoon rainfall.

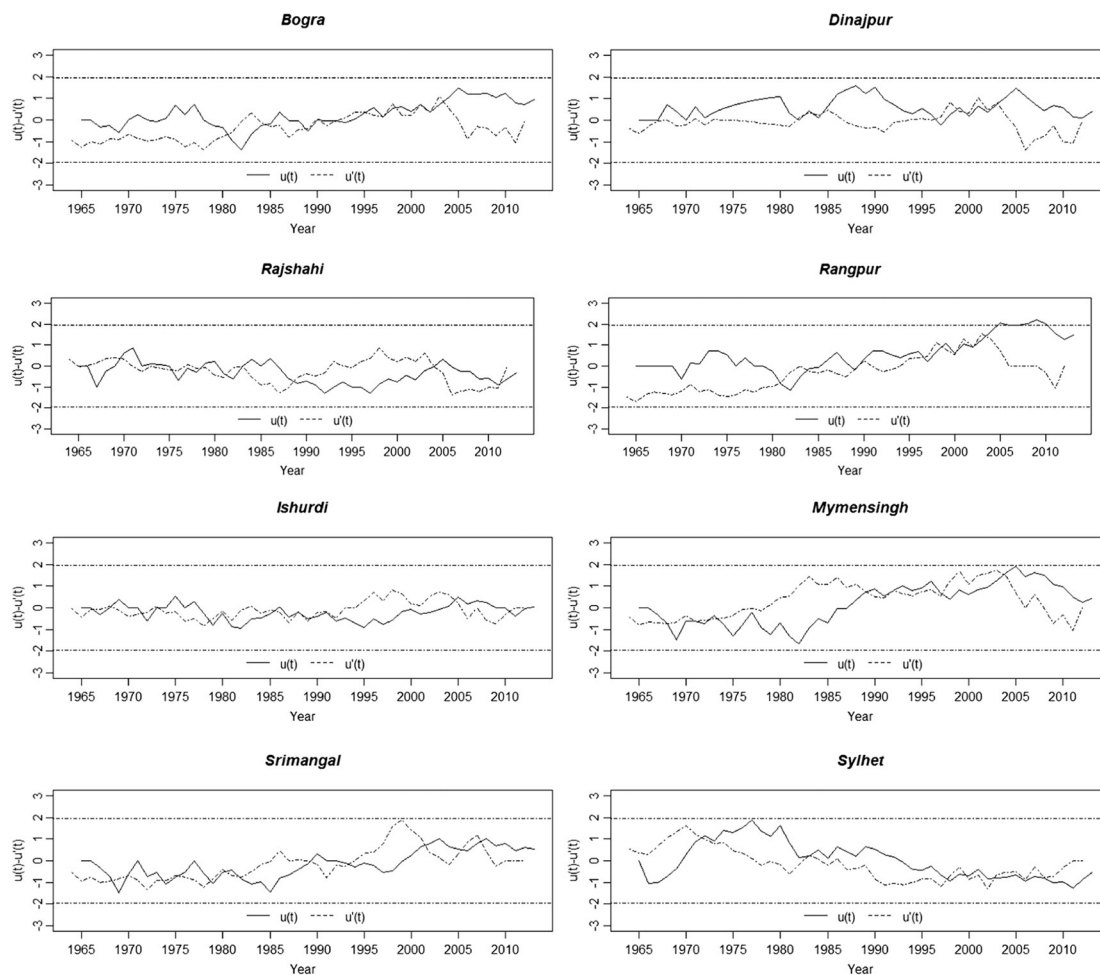


Fig. 9. Sequential Mann-Kendall test rank statistics for post-monsoon rainfall. Dotted horizontal straight lines indicate the lower limit and upper limit of 95% confidence interval.

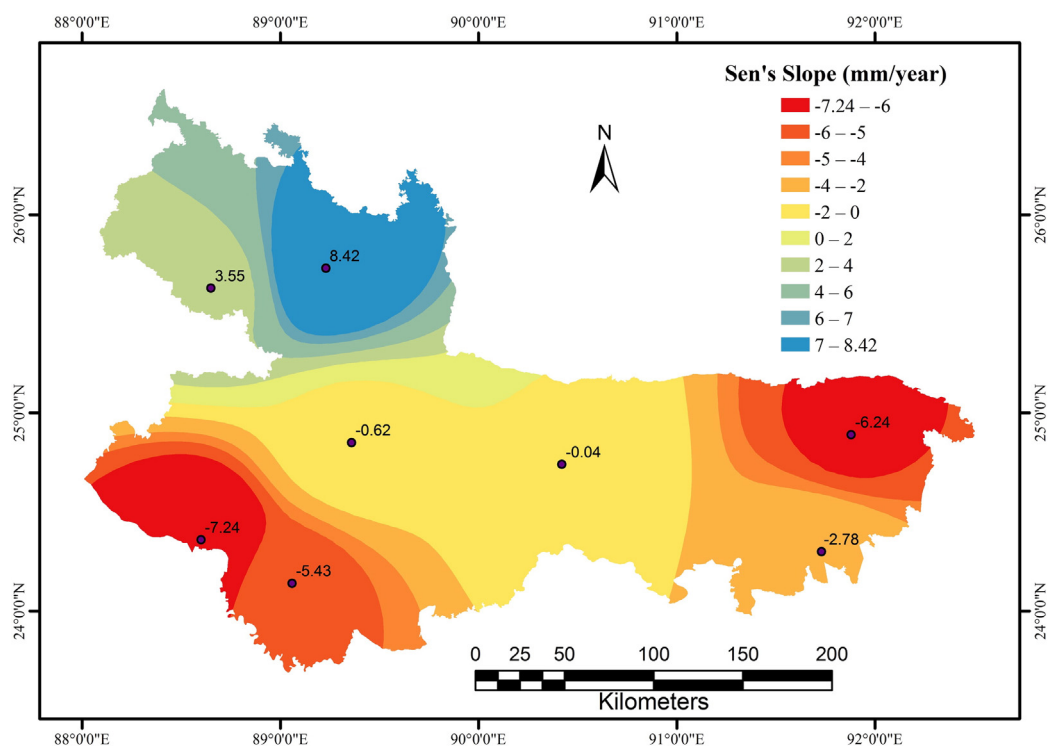


Fig. 10. Spatial distribution of trend magnitude (mm/year) for annual rainfall.

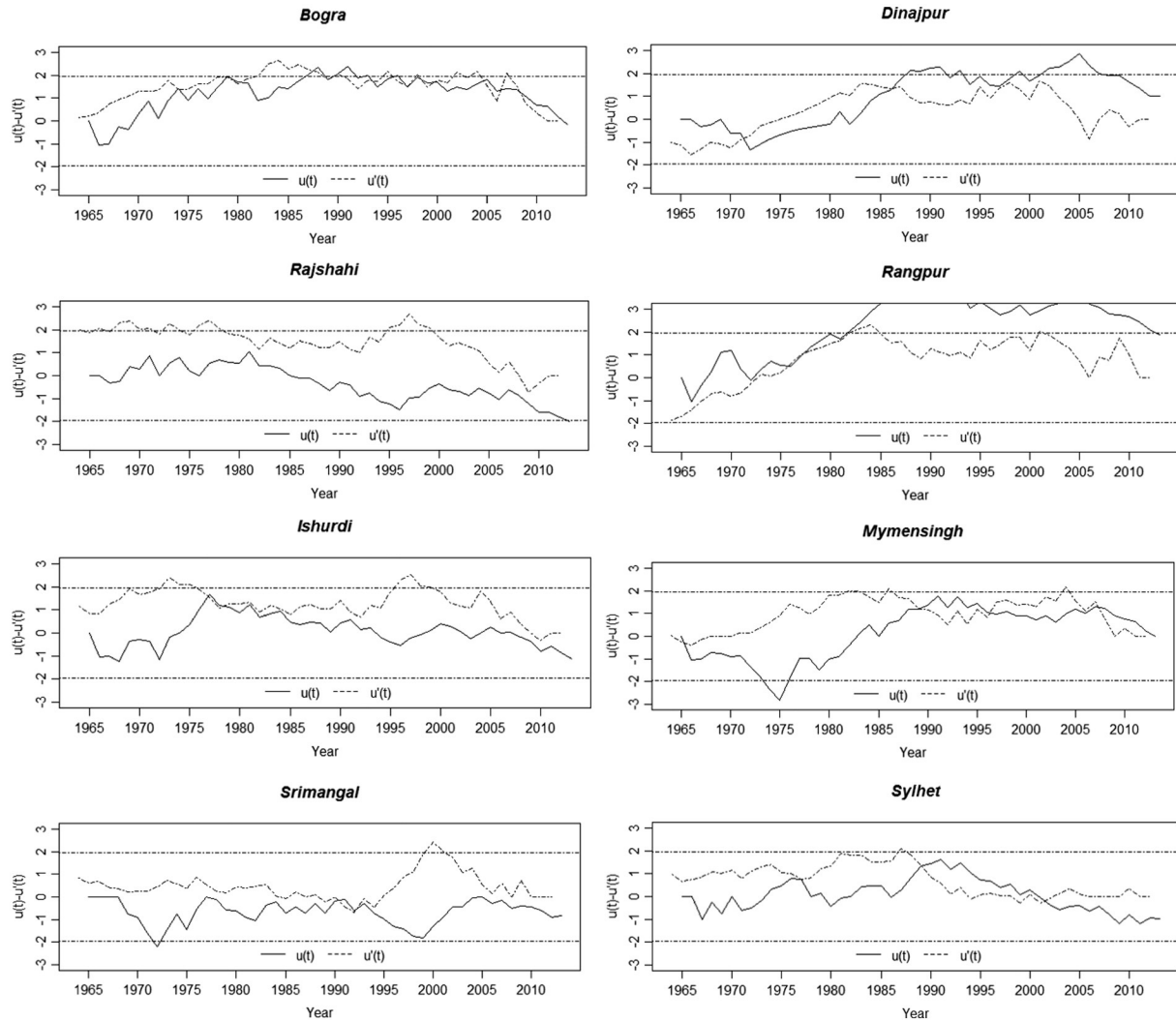


Fig. 11. Sequential Mann-Kendall test rank statistics for annual rainfall. Dotted horizontal straight lines indicate the lower limit and upper limit of 95% confidence interval.

et al. (2001) found a weakening of the monsoon circulation using results from the recent atmospheric general circulation experiments for Asian summer monsoon. The decrement in south Asian monsoon was also found in other studies (Loo et al., 2014; Roxy et al., 2015; Schewe and Levermann, 2012; Sinha et al., 2015).

Monsoon in Bangladesh is linked with El Niño Southern Oscillation (ENSO) (Ahasan et al., 2010). During the year of El Niño, an anomalous subsidence suppresses convection over South Asia and results in a weaker monsoon. However, ENSO connection with monsoon rainfall needs a further detailed study for a concrete conclusion. Another possible cause of decreasing monsoon rainfall trend may be the rapid warming in the Indian Ocean. Warming of Indian Ocean potentially weakens the land-sea thermal difference that reduces the rainfall over parts of South Asia (Roxy et al., 2015).

Although the Mann-Kendall test reveals a positive trend in pre-monsoon and post-monsoon rainfall and a negative trend in the monsoon, their monotonic relation with time were weak. Hence, sequential Mann-Kendall test was applied to see if there were any abrupt changes. Results show a periodic fluctuation of rainfall trend in most cases. Analysis of longer periods of data would have revealed if there were any cyclic patterns in seasonal rainfall in northern Bangladesh. In addition, the impact of global climate change and changes in local physiography on rainfall behaviour need to be further explored.

8. Conclusion

This study analyses seasonal and annual rainfall temporal trends and their fluctuations over time. The analysis covers the five distinguished climatic subdivisions of the country. The non-parametric Mann-Kendall test shows that winter rainfall has an increasing trend in northeast climatic subdivision whereas the negative trend prevails in the northern part of the northern region. However, pre-monsoon rainfall tends to increase except at Ishurdi station. Monsoon rainfall is found to be decreasing in most of northern Bangladesh with the exceptions at Dinajpur and Rangpur station. Dinajpur has a neutral monsoon rainfall, whereas monsoon rainfall at Rangpur station was found increasing at a rate of 2.02 mm/year. Annual rainfall exhibits similar trend as monsoon. Post-monsoon rainfall shows a positive trend in the majority of the locations. However, seasonal and annual changes were found statistically non-significant at 95% confidence limit for all stations but monsoon and annual rainfall at Rajshahi station. The decreasing trend of monsoon and annual rainfall at Rajshahi was found significant at 95% confidence limit. The sequential Mann-Kendall test results were interpreted as periodic fluctuations of seasonal and annual rainfall in the area although a continuous decreasing trend was found in few cases. However, after the year 1990, a non-significant decreasing trend in most cases was identified. It is anticipated that the findings of the current study will have a

substantial effect on water resources and agricultural planning in this area. Agriculture in Bangladesh is highly dependent on natural rainfall. Any changes in rainfall amount, especially change in seasonal rainfall, may change the physiology of plants. Shortage of rainfall in irrigation time may cause stress to ground water extraction. Cropping pattern may require changes due to temporal shift in rainfall.

Acknowledgement

We thank ASK Scientific for helping in language editing. The author, SHB would like to thank Ministry of Science and Technology, Bangladesh for awarding a MSc. fellowship. MTR and MAH extend their gratitude to Ministry of Science and Technology, Bangladesh for providing Prime Minister's Special Allocation to conduct research at SUST. The authors are very much grateful to anonymous reviewers whose comments have helped to improve the quality of this manuscript.

References

- Afzal, M., Mansell, M.G., Gagnon, A.S., 2011. Trends and variability in daily precipitation in Scotland. *Procedia Environ. Sci.* 6, 15–26. <http://dx.doi.org/10.1016/j.proenv.2011.05.003>.
- Agrawala, S., Ota, T., Ahmed, A.U., Smith, J., Van Aalst, M., 2003. Development and Climate Change in Bangladesh: Focus on Coastal Flooding and the Sundarbans (OECD Fr).
- Ahmed, F., Hewa, G.A., Argue, J.R., 2014. Variability of annual daily maximum rainfall of Dhaka, Bangladesh. *Atmos. Res.* 137, 176–182. <http://dx.doi.org/10.1016/j.atmosres.2013.10.013>.
- Ahasan, M.N., Chowdhury, M.A., Quadri, D.A., 2010. Variability and trends of summer monsoon rainfall over Bangladesh. *J. Hydrol. Meteorol.* 7, 1–17.
- Ahmed, R., 1994. Variabilities of rainfall and duration of the summer monsoon, and their relationships with monsoon onset dates in Bangladesh. *Proc. Intl. Conf. on Monsoon Prediction and Variability*, pp. 320–327.
- Ahmed, R., Karmakar, S., 1993. Arrival and withdrawal dates of the summer monsoon in Bangladesh. *Int. J. Climatol.* 13, 727–740. <http://dx.doi.org/10.1002/joc.3370130703>.
- Ahmed, R., Kim, I.-K., 2003. Patterns of daily rainfall in Bangladesh during the summer monsoon season: case studies at three stations. *Phys. Geogr.* 24, 295–318. <http://dx.doi.org/10.2747/0272-3646.24.4.295>.
- Basistha, A., Arya, D.S., Goel, N.K., 2009. Analysis of historical changes in rainfall in the Indian Himalayas. *Int. J. Climatol.* 29, 555–572. <http://dx.doi.org/10.1002/joc.1706>.
- Briffa, K.R., van der Schrier, G., Jones, P.D., 2009. Wet and dry summers in Europe since 1750: evidence of increasing drought. *Int. J. Climatol.* 29, 1894–1905. <http://dx.doi.org/10.1002/joc.1836>.
- Brooker, P.I., 1991. *A Geostatistical Primer*. World Scientific.
- Choudhury, B.U., Das, A., Ngachan, S.V., Slong, A., Bordoloi, L.J., Chowdhury, P., 2012. Trend analysis of long term weather variables in mid altitude Meghalaya, North-East India. *J. Agric. Phys.* 12, 12–22.
- Choudhury, S.A., Terao, T., Murata, F., Hayashi, T., 2013. Seasonal variations of temperature and rainfall characteristics in the northeastern part of Bangladesh around Sylhet. *J. Agrofor. Env.* 6.
- Farhana, S., Rahman, M.M., 2011. Characterizing rainfall trend in bangladesh by temporal statistics analysis. 4th Annual Paper Meet and 1st Civil Engineering Congress, Dhaka, Bangladesh.
- Fortin, M.-J., Dale, M.R.T., 2005. *Spatial Analysis: A Guide for Ecologists*. Cambridge University Press.
- Hamed, K.H., Rao, A.R., 1998. A modified Mann–Kendall trend test for autocorrelated data. *J. Hydrol.* 204, 182–196. [http://dx.doi.org/10.1016/S0022-1694\(97\)00125-X](http://dx.doi.org/10.1016/S0022-1694(97)00125-X).
- Hasan, Z., Akhter, S., Islam, M., 2014. Climate change and trend of rainfall in the south-east part of coastal Bangladesh. *Eur. Sci. J.* 10.
- Hirsch, R.M., Helsel, D.R., Cohn, T.A., Gilroy, E.J., Maidment, D.R., others, 1992. *Statistical analysis of hydrologic data. Handbook of Hydrology* (17–1).
- Immerzeel, W., 2008. Historical trends and future predictions of climate variability in the Brahmaputra basin. *Int. J. Climatol.* 28, 243–254. <http://dx.doi.org/10.1002/joc.1528>.
- Islam, M.N., Uyeda, H., 2008. Vertical variations of rain intensity in different rainy periods in and around Bangladesh derived from TRMM observations. *Int. J. Climatol.* 28, 273–279. <http://dx.doi.org/10.1002/joc.1585>.
- Jha, M.K., Singh, A.K., 2013. Trend analysis of extreme runoff events in major river basins of Peninsular Malaysia. *Int. J. Water* 7, 142. <http://dx.doi.org/10.1504/IJW.2013.051995>.
- Karimi, H.A., 2009. *Handbook of Research on Geoinformatics*. IGI Global.
- Karmakar, S., Quadri, D.A., Mannan, M.A., 2015. Trends in maximum temperature and thunderstorms, their correlation and impacts on the livelihood of Bangladesh. *Atmosphere* 5.
- Kendall, M.G., 1938. A new measure of rank correlation. *Biometrika* 30, 81–93.
- Kendall, M.G., 1948. *Rank Correlation Methods*, First edition. ed. C. Griffin.
- Kripalani, R.H., Oh, J.H., Kulkarni, A., Sabade, S.S., Chaudhari, H.S., 2007. South Asian summer monsoon precipitation variability: coupled climate model simulations and projections under IPCC AR4. *Theor. Appl. Climatol.* 90, 133–159. <http://dx.doi.org/10.1007/s00704-006-0282-0>.
- Lloyd, C., 2010. *Spatial Data Analysis: An Introduction for GIS Users*. OUP Oxford.
- Longley, P., 2005. *Geographic Information Systems and Science*. John Wiley & Sons.
- Loo, Y.Y., Billa, L., Singh, A., 2014. Effect of climate change on seasonal monsoon in Asia and its impact on the variability of monsoon rainfall in Southeast Asia. *Geosci. Front.* - <http://dx.doi.org/10.1016/j.gsf.2014.02.009>.
- Lu, A., He, Y., Zhang, Z., Pang, H., Gu, J., 2004. Regional structure of global warming across China during the twentieth century. *Clim. Res.* 27, 189–195.
- Mann, H.B., 1945. Nonparametric tests against trend. *Econ. J. Econ. Soc.* 13, 245–259.
- Martinez, C.J., Maleski, J.J., Miller, M.F., 2012. Trends in precipitation and temperature in Florida, {USA}. *J. Hydrol.* 452–453, 259–281. <http://dx.doi.org/10.1016/j.jhydrol.2012.05.066>.
- McCarthy, J.J., 2001. Climate change 2001: impacts, adaptation, and vulnerability: contribution of Working Group II to the third assessment report of the Intergovernmental Panel on Climate Change. Cambridge University Press.
- Mosmann, V., Castro, A., Fraile, R., Dessens, J., Sánchez, J., 2004. Detection of statistically significant trends in the summer precipitation of mainland Spain. *Atmos. Res.* 70, 43–53. <http://dx.doi.org/10.1016/j.atmosres.2003.11.002>.
- Nasri, M., Modarres, R., 2009. Dry spell trend analysis of Isfahan Province, Iran. *Int. J. Climatol.* 29, 1430–1438. <http://dx.doi.org/10.1002/joc.1805>.
- Nowreen, S., Murshed, S.B., Islam, A.K.M.S., Bhaskaran, B., Hasan, M.A., 2015. Changes of rainfall extremes around the haor basin areas of Bangladesh using multi-member ensemble RCM. *Theor. Appl. Climatol.* 119, 363–377. <http://dx.doi.org/10.1007/s00704-014-1101-7>.
- Partal, T., Kahya, E., 2006. Trend analysis in Turkish precipitation data. *Hydrol. Process.* 20, 2011–2026. <http://dx.doi.org/10.1002/hyp.5993>.
- Patra, J., Mishra, A., Singh, R., Raghuvanshi, N.S., 2012. Detecting rainfall trends in twentieth century (1871–2006) over Orissa State, India. *Clim. Chang.* 111, 801–817. <http://dx.doi.org/10.1007/s10584-011-0215-5>.
- Rahman, A., Haque Jiban, J., Munna, S.A., 2015. Regional variation of temperature and rainfall in Bangladesh: estimation of trend. *Open J. Stat.* 05, 652–657. <http://dx.doi.org/10.4236/ojs.2015.7066>.
- Rashid, H.E., 1991. *Geography of Bangladesh*. Second. ed. The University Press Limited, Dhaka.
- Roxy, M.K., Ritika, K., Terray, P., Murtugudde, R., Ashok, K., Goswami, B.N., 2015. Drying of Indian subcontinent by rapid Indian Ocean warming and a weakening land-sea thermal gradient. *Nat. Commun.* 6, 7423. <http://dx.doi.org/10.1038/ncomms8423>.
- Sanderson, M., Ahmed, R., 1979. Pre-monsoon rainfall and its variability in Bangladesh: a trend surface analysis/Chute des pluies de la pré-mousson et sa variabilité au Bangladesh: Tendence d'analyse en superficie. *Hydrol. Sci. Bull.* 24, 277–287. <http://dx.doi.org/10.1080/02626667909491867>.
- Schewe, J., Levermann, A., 2012. A statistically predictive model for future monsoon failure in India. *Environ. Res. Lett.* 7, 044023. <http://dx.doi.org/10.1088/1748-9326/7/4/044023>.
- Sen, P.K., 1968. Estimates of the regression coefficient based on Kendall's tau. *J. Am. Stat. Assoc.* 63, 1379–1389. <http://dx.doi.org/10.1080/01621459.1968.10480934>.
- Shahid, S., 2008. Spatial and temporal characteristics of droughts in the western part of Bangladesh. *Hydrol. Process.* 22, 2235–2247. <http://dx.doi.org/10.1002/hyp.6820>.
- Shahid, S., 2010a. Rainfall variability and the trends of wet and dry periods in Bangladesh. *Int. J. Climatol.* 30, 2299–2313. <http://dx.doi.org/10.1002/joc.2053>.
- Shahid, S., 2010b. Recent trends in the climate of Bangladesh. *Clim. Res.* 42, 185–193. <http://dx.doi.org/10.3354/cr00889>.
- Shahid, S., 2011. Trends in extreme rainfall events of Bangladesh. *Theor. Appl. Climatol.* 104, 489–499. <http://dx.doi.org/10.1007/s00704-010-0363-y>.
- Shahid, S., Behrawan, H., 2008. Drought risk assessment in the western part of Bangladesh. *Nat. Hazards* 46, 391–413. <http://dx.doi.org/10.1007/s11069-007-9191-5>.
- Shahid, S., Khairulmaini, O.S., 2009. Spatio-temporal variability of rainfall over Bangladesh during the time period 1969–2003. *Asia-Pac. J. Atmos. Sci.* 45, 375–389.
- Siddik, M.A., Rahman, M., 2014. Trend analysis of maximum, minimum, and average temperatures in Bangladesh: 1961–2008. *Theor. Appl. Climatol.* 116, 721–730. <http://dx.doi.org/10.1007/s00704-014-1135-x>.
- Sinha, A., Kathayat, G., Cheng, H., Breitenbach, S.F.M., Berkelhammer, M., Mudelsee, M., Biswas, J., Edwards, R.L., 2015. Trends and oscillations in the Indian summer monsoon rainfall over the last two millennia. *Nat. Commun.* 6, 6309. <http://dx.doi.org/10.1038/ncomms7309>.
- Some'e, B.S., Ezani, A., Tabari, H., 2012. Spatiotemporal trends and change point of precipitation in Iran. *Atmos. Res.* 113, 1–12. <http://dx.doi.org/10.1016/j.atmosres.2012.04.016>.
- Sonali, P., Kumar, D.N., 2013. Review of trend detection methods and their application to detect temperature changes in India. *J. Hydrol.* 476, 212–227. <http://dx.doi.org/10.1016/j.jhydrol.2012.10.034>.
- Stephenson B., Douville, H., Kumar, K.R., 2001. Searching for a Fingerprint of Global Warming in the Asian Summer Monsoon. *MAUSAM* 52, 213–220.
- Tabari, H., Hosseinzadeh Talaei, P., Ezani, A., Shifteh Some'e, B., 2012. Shift changes and monotonic trends in autocorrelated temperature series over Iran. *Theor. Appl. Climatol.* 109, 95–108. <http://dx.doi.org/10.1007/s00704-011-0568-8>.
- Trenberth, K.E., 2011. Changes in precipitation with climate change. *Clim. Res.* 47, 123–138. <http://dx.doi.org/10.3354/cr00953>.
- Vasilades, L., Loukas, A., Patsonas, G., 2009. Evaluation of a statistical downscaling procedure for the estimation of climate change impacts on droughts. *Nat. Hazards Earth Syst. Sci.* 9, 879–894. <http://dx.doi.org/10.5194/nhess-9-879-2009>.
- Wang, W., Bobojonov, I., Härdle, W.K., Odening, M., 2013. Testing for increasing weather risk. *Stoch. Env. Risk A.* 27, 1565–1574. <http://dx.doi.org/10.1007/s00477-013-0692-3>.
- Yamane, Y., Hayashi, T., Kiguchi, M., Akter, F., Dewan, A.M., 2013. Synoptic situations of severe local convective storms during the pre-monsoon season in Bangladesh. *Int. J. Climatol.* 33, 725–734. <http://dx.doi.org/10.1002/joc.3460>.
- Zhang, L., Zhou, T., 2011. An assessment of monsoon precipitation changes during 1901–2001. *Clim. Dyn.* 37, 279–296. <http://dx.doi.org/10.1007/s00382-011-0993-5>.
- Zhang, Q., Xu, C.-Y., Zhang, Z., 2009. Observed changes of drought/wetness episodes in the Pearl River basin, China, using the standardized precipitation index and aridity index. *Theor. Appl. Climatol.* 98, 89–99. <http://dx.doi.org/10.1007/s00704-008-0095-4>.